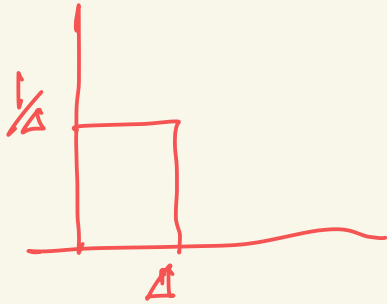




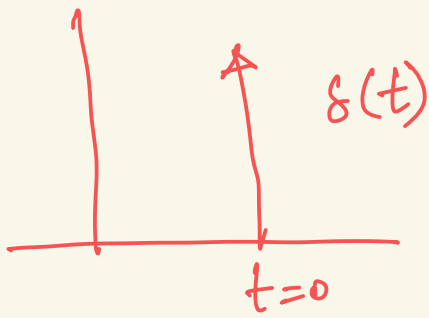
Lecture 13

Impulse function:

Let's say there is a signal $p(t)$ whose area is 1. Its width is Δ and height is $\frac{1}{\Delta}$



Now if we reduce Δ , then the height $\frac{1}{\Delta}$ increases. That is, $\Delta \rightarrow 0$, $\frac{1}{\Delta} \rightarrow \infty$. And we write the function in the following way.

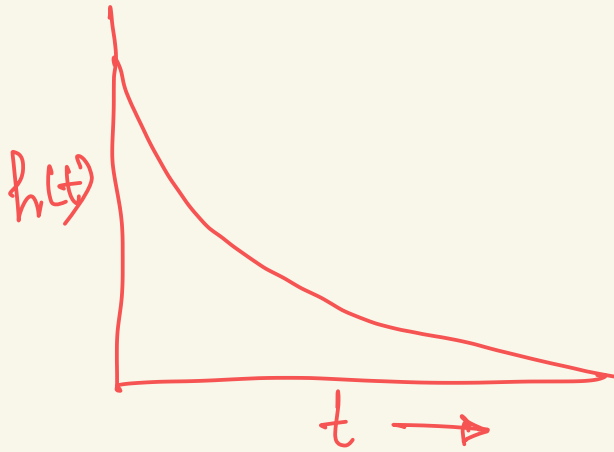


This $\delta(t)$ function has height infinity and thickness zero. This type of function is called impulse.

This type of function is used to model forces which has very high magnitude within a very short time.

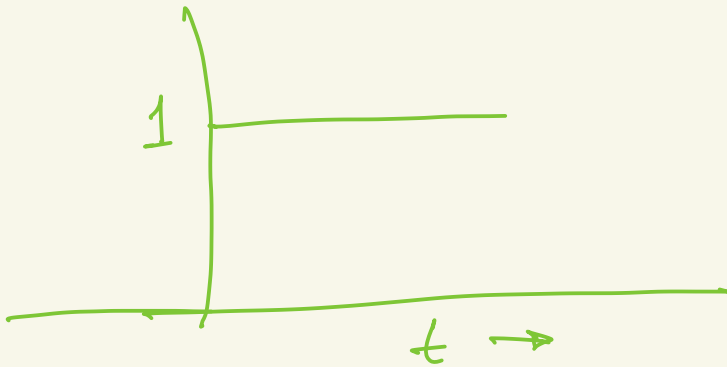
$$\begin{aligned}\delta(t) &= \infty, & t &= 0 \\ &= 0, & t &\neq 0\end{aligned}$$

the response of a system due to this impulse is called the impulse response. A typical impulse response looks the following



$h(t)$ is the impulse response. A transfer function is basically the impulse response of a system in the Laplace domain.

Step function



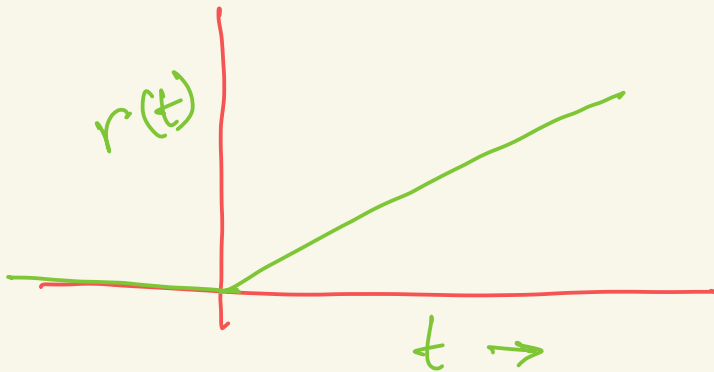
A step function has a constant value of one at $t \geq 0$. The value of the step function is zero when $t < 0$.

$$f(t) = 1, \quad t \geq 0$$
$$= 0, \quad t < 0$$

Ramp function

A ramp function is basically a

straight line passing through zero. It can be defined analytically in the following way.



$$\begin{aligned} r(t) &= t, \quad t \geq 0 \\ &= 0, \quad t < 0 \end{aligned}$$

Sifting property:

If $f(t)$ is a continuous function at $t = \tau$, then we can

find the value of the following integral by using the sifting property

$$\int_{-\infty}^{\infty} f(\gamma) \delta(t-\gamma) d\gamma = f(t) \quad \text{--- (1)}$$

Convolution:

if we replace f by u in equation 1, then it represents an input $u(t)$ as a sum of impulses of intensity $u(t-\gamma)$.

To find the response of an

arbitrary input, the principle of superposition tells us that we only need to find the response to a unit impulse.

If the system is not only linear but also time invariant, then the impulse response is given by $h(t - \tau)$

And the output for a general input is given by

that integral.

$$y(t) = \int_{-\infty}^{\infty} u(\tau) h(t-\tau) d\tau$$

By changing variables as

$$\tau_1 = t - \tau$$

$$y(t) = \int_{\infty}^{-\infty} u(t-\tau_1) h(\tau_1) (-d\tau_1)$$

$$y(t) = \int_{-\infty}^{\infty} h(\tau) u(t-\tau) d\tau$$

This integral is called convolution.

In the Laplace domain the integral becomes a multiplication

$$Y(s) = V(s)H(s)$$

As a result, it is easy to find the convolution in the Laplace domain.